

Initialization

```
In[1]:= SetDirectory[NotebookDirectory[]];
NotebookEvaluate[FileNameJoin[{NotebookDirectory[], "helper_functions.nb"}]];
NotebookEvaluate[FileNameJoin[{NotebookDirectory[], "dls_mem_functions.nb"}]];
```

Data import

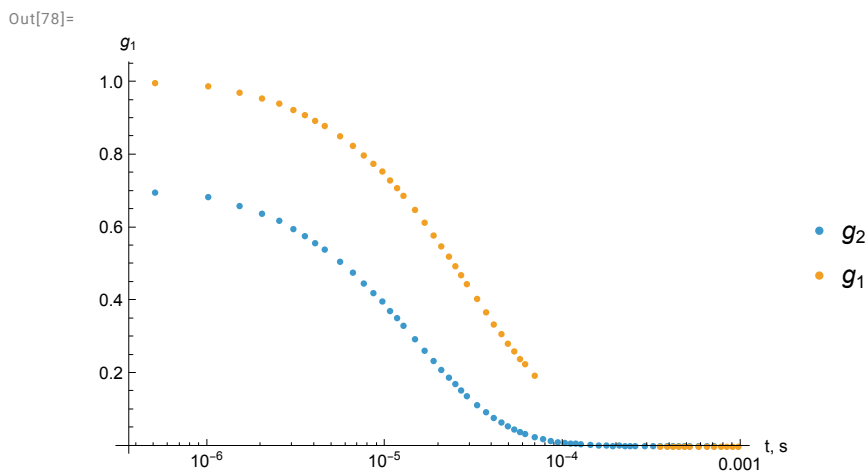
The file *dls_mass_phot_bsa.txt* contains the dls data for 3 mg/mL BSA in HEPES buffer at pH 7.0.

```
In[4]:= fnd = FileNames["dls*.txt"];
TableForm[%]
```

```
Out[5]//TableForm=
dls_mass_phot_bsa.txt
```

The plot below compares the original g_2 data from the Malvern instrument (blue) with the processed g_1 data generated using clipG2toG1.

```
In[76]:= dbsa = N@{10-6 dls[m[fnd[[1]]][[4, 1]], dls[m[fnd[[1]]][[5, 1]]}^T;
g1bsac = clipG2toG1[dbsa, 16 × 10-2, 1 × 10-4, 13];
ListPlot[{Take[dbsa, 60], g1bsac}, ScalingFunctions → {"Log"},
PlotRange → All, PlotLegends → {"g2", "g1"}, AxesLabel → {"t, s", "g1"}]
```



Feasibility boundary constant C_b for BSA, determined experimentally as the value at which dlsMEMc converges.

```
In[79]:= cbsab = 0.0024;
```

Running constrained MEM for two values of C_b .

```
In[83]:= Timing[outc = Quiet[dlsMEMc[g1bsac, 0.02, (# cbsab)] & /@ {1., 2.}]];
```

```
Out[83]=
{31.9581, Null}
```

Recovered distributions for two values of C_b .

```

In[82]:= ListLinePlot[{{#[[1]]-1,  $\frac{\#[[2]]}{\text{Max}[\#[[2]]]}$ }T & /@ outc,
  ScalingFunctions → {"Log"}, PlotRange → {{ $0.1 \times 10^{-5}$ ,  $0.6 \times 10^{-2}$ }, All},
  AxesLabel → {"t, s", "normalized probability"}, PlotLegends → {"1x Cb", "2x Cb"}]

```

Out[82]=

